

Decarbonization

**The Greatest Environmental Engineering Problem
of Our Time**

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ABOUT ME

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EDUCATION BACKGROUND

BEIJING NORMAL UNIVERSITY

B.Sc. · Environmental Science



PRINCETON UNIVERSITY

Ph.D. · Public Policy



HARVARD UNIVERSITY

Postdoctoral Fellow · Environmental Policy



MISSION

Bridging engineering and technical analysis with **policy and economic thinking**

RESEARCH FOCUS

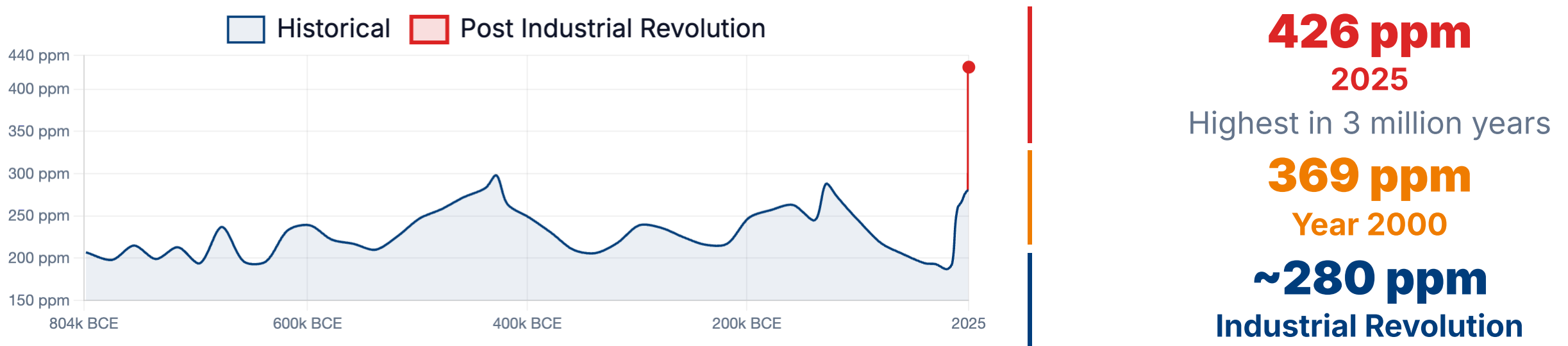
Decarbonize the whole economy

How do we decarbonize the whole economy to reach net zero?

Impact of climate change on our world

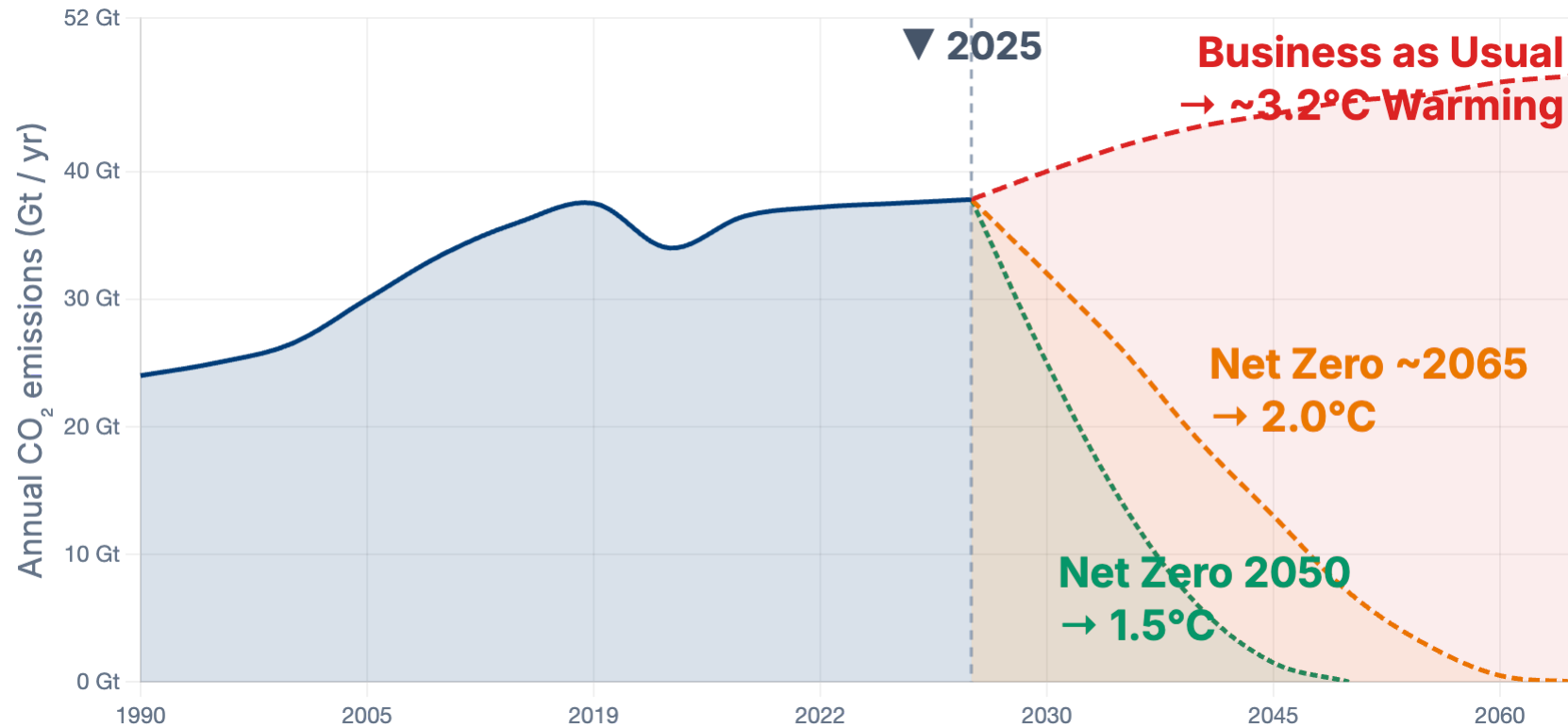
Understanding how a changing climate reshapes societies, economies, and ecosystems.

Atmospheric CO₂ Concentration over the Past 800,000 Years



Rapid rise since 1760 (Industrial Revolution) is largely driven by a **fossil fuel economy** — burning coal, oil, and gas releases CO₂ stored underground for millions of years.

Global CO₂ Emissions: Where We Need to Go



Source: Global Carbon Project · IPCC AR6 (2022)

TOTAL EMISSIONS (2025)

🇨🇳 China	32 %
🇺🇸 USA	13 %
🇮🇳 India	8 %
🇪🇺 EU27 + UK	6 %

PER CAPITA (T/PERSON, 2023)

🇺🇸 USA	13.8 t
🇨🇳 China	9.2 t
🇸🇬 Singapore	8.9 t
🇪🇺 EU	5.7 t
🇮🇳 India	2.1 t

ONE OF THE GREATEST CHALLENGES OF OUR TIME

How do we decarbonize the global economy?

We must reduce global CO₂ emissions from ~38
Gt/yr today to **net zero** within your working
lifetime — while powering a growing world
economy.

The Net Zero Timeline

2030

Emissions must halve

SG carbon tax rising ·
EVs mainstream ·
Renewables dominant

2050

**EU · USA ·
Singapore net
zero**

Entire energy
infrastructure rebuilt

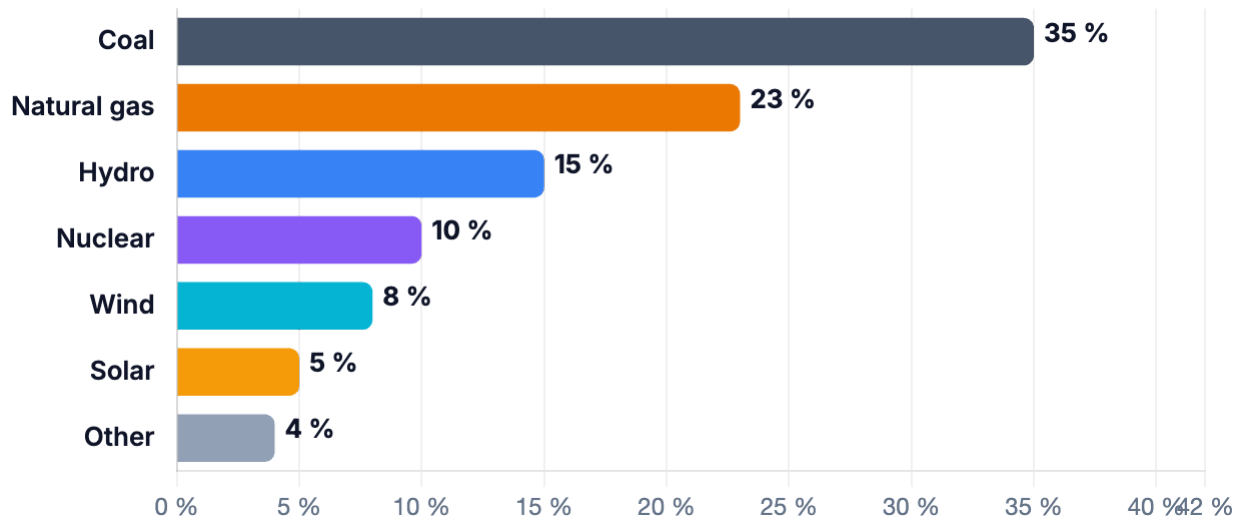
2060– 70

**China · India ·
Global net zero**

IPCC outer boundary · Your
senior career

Where Does Electricity Come From Today?

GLOBAL ELECTRICITY MIX (2023)



Source: IEA 2023

SINGAPORE ELECTRICITY MIX (2023)



Singapore runs almost entirely on **fossil gas** — almost no hydro, wind, or nuclear options. Solar and regional imports are critical.

The Clean Alternative: Wind + Solar + Storage



Solar PV

Now the **cheapest** source of new electricity in most of the world



Wind

Onshore wind cheaper than gas in most regions. Offshore wind expanding rapidly.

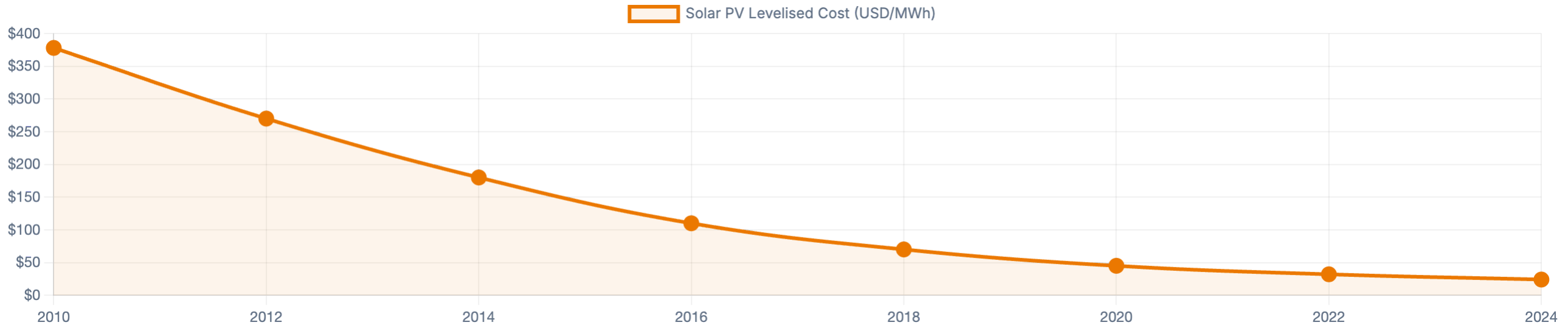


Storage

Batteries store excess solar/wind and release it when the sun is not shining

Together, these three can replace fossil fuel power — **but the grid must be completely redesigned.**

Solar Energy: 99 % cheaper in 40 years



-99 %

Solar PV module cost since 1976. From \$106/W → ~\$0.2/W today.

Why does it still need to fall further?

Storage, grid integration, and land costs remain high. Total system cost — not just the panel — determines competitiveness.

The Energy Trilemma

Every country faces a trade-off between three goals — you cannot fully optimise all three at once.



Security

Is the power always on?
Can we survive supply
disruptions?



Affordability

Can households and
industry afford it? Does it
harm economic growth?



Sustainability

Is it low-carbon? Does it
protect air, water, and
land?

Designing a net zero energy system means **navigating this trilemma** for every country, every city, every grid. This is the core of environmental systems engineering.

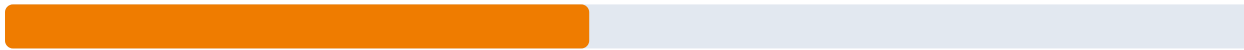
Electric Vehicles: How Much Cleaner?

Lifecycle CO₂ per km — including manufacturing & electricity generation:

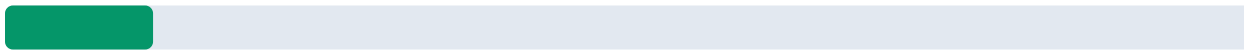
Petrol / Diesel car **~170 g CO₂/km**



EV on average global grid **~80 g CO₂/km**



EV on clean grid **~20 g CO₂/km**

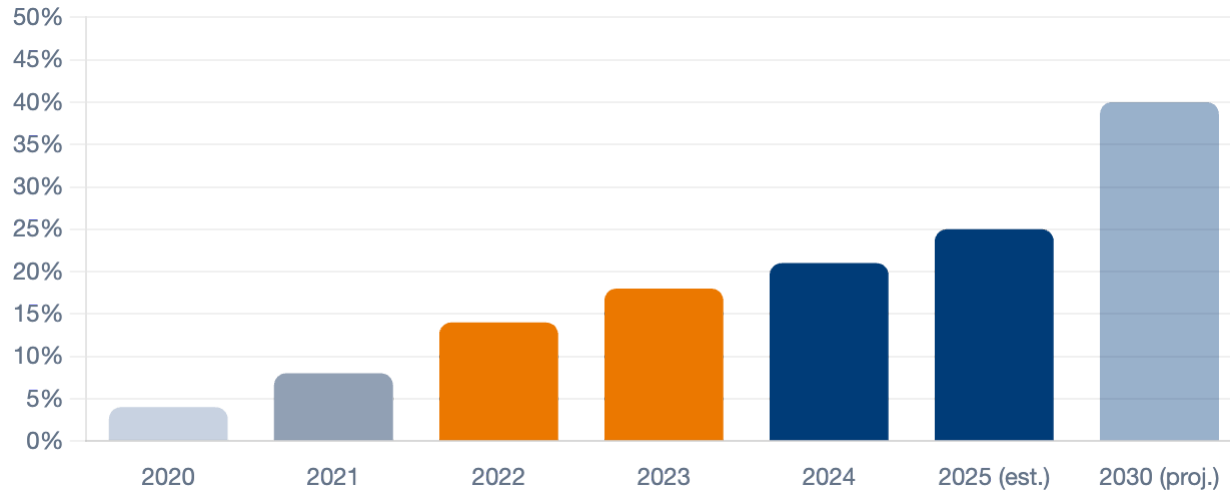


Key insight: The cleaner the grid, the cleaner the EV. Decarbonizing power and transport are *linked*.

Singapore: EVs today run mostly on natural gas. As solar grows, every EV automatically gets cleaner — no hardware change needed.

The EV Revolution Is Already Here

GLOBAL EV SHARE OF NEW CAR SALES (%)



China

World's largest EV market. Over 40 % of new cars sold are EVs (2024).

Europe

Many countries targeting 100 % EV new sales by 2035.

Singapore

Phase out new ICE registrations by **2030**.
All vehicles on cleaner energy by **2040**.

Singapore: Hub of Global Shipping

#1

Largest bunkering
port in the world

3,000+

Ships refuel every
month

\$50B+

Annual bunkering revenue

The Strait of Malacca — one of the world's busiest waterways — runs through Singapore's backyard. Over **80,000 vessels** transit it each year.

Singapore's maritime sector contributes **~7 % of GDP**. Shipping decarbonization is not just an environmental issue — it is an **economic survival question**.

A NUMBER THAT SHOULD SURPRISE YOU

40 %

of global shipping tonnage is **fossil fuel**.

Oil tankers, LNG carriers, coal bulk carriers — nearly half of everything on the world's oceans is the fuel we are trying to stop burning. Nearly every ship runs on **heavy fuel oil**.

If shipping decarbonises, Singapore's entire bunkering economy must transform.
IMO target: net zero shipping by 2050.

Decarbonizing Shipping: The Hard Problem

Ships cannot be easily electrified — they need energy-dense fuels for long ocean voyages.

Fuel	Status	Key issue
LNG	Available	Still a fossil fuel
Ammonia	Pilot stage	Toxic handling
Methanol	Early use	Green supply scarce
Green H ₂	Emerging	Cost — still 3–5× grey H ₂

Singapore's bet:

Become a **green hydrogen and ammonia bunkering hub** — the world's first green fuel station for ships.

The research challenge:

Designing storage, safety systems, supply chains, and port infrastructure for fuels that don't yet exist at scale. Open engineering territory.

QUICK QUESTION

Which of these has the highest carbon footprint per kilogram of food?



Beef



Chicken

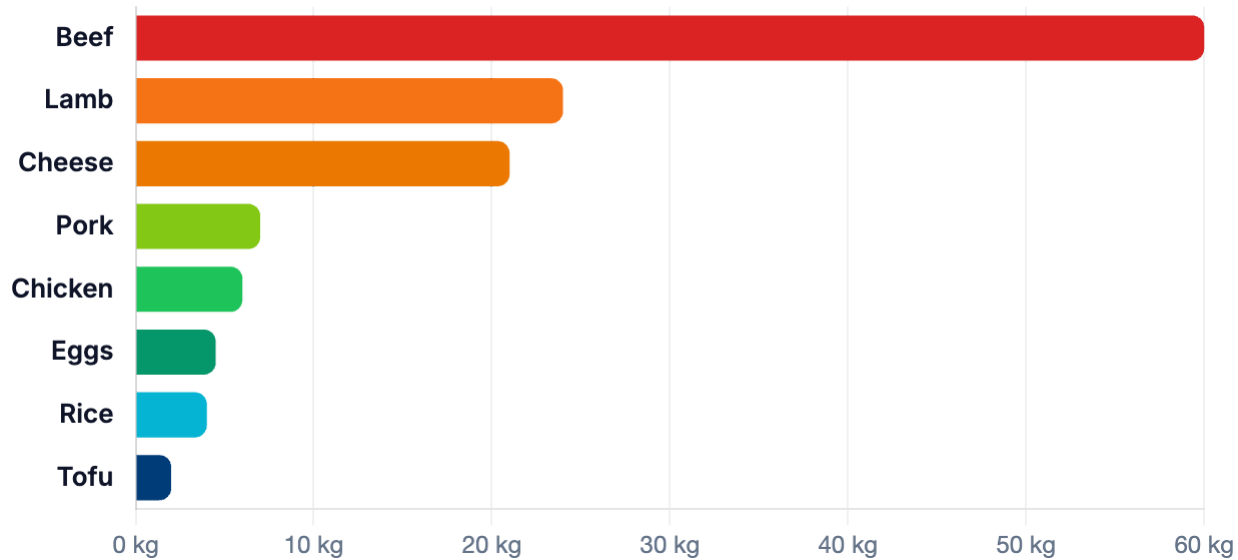


Rice



Vegetables

The Answer: Beef



Source: Our World in Data / Poore & Nemecek (2018)

60 kg CO₂
per kg of beef produced

Why so much?

Cattle release **methane** through digestion — 80× more potent than CO₂. Add land clearing, feed crops, and water use.

Food systems = **26 % of global GHG emissions**.
What we eat is a decarbonization problem.

We Need a New Framework

Designing a decarbonization pathway means balancing **five competing dimensions simultaneously**:

Timing

When must emissions fall? What is feasible by 2030, 2050, 2070?

Cost

Who pays? What technologies are affordable now vs. in 10 years?

Energy security

Can we keep the lights on during the transition?

Land, water & air

Solar needs land. Wind affects ecosystems. Every solution has side-effects.

No single technology solves this. It requires **systems thinking across all dimensions** — exactly what environmental and systems engineers are trained to do.

THE OPPORTUNITY

Decarbonization is one of the fastest-growing industries of the 21st century

24M

new green jobs
by 2030
— ILO

\$1.8T

clean energy
investment in
2023
— IEA

#1

most active VC
sector globally
Climate tech

What Can You Do With This Degree?

Energy Systems Engineer

Design power grids, storage systems, and microgrids for a renewables-dominant world

Energy Market Analyst

Carbon markets, electricity pricing, policy modelling — data-driven decision making

ESG Consultant

Help companies measure and reduce emissions — now mandatory for large firms globally

Policy Advisor

Singapore, ASEAN, UN — translating engineering reality into workable regulation

Climate Tech Founder

Build the startups solving problems that the world's largest companies cannot

Researcher

Open questions remain in storage, hydrogen, carbon removal — NUS is at the frontier

ONE QUESTION

**In 2070, when the world looks back
at how this problem was — or wasn't — solved,
where do you want to have been?**

The decision you make in the next few weeks is one small step in answering that question.

Questions?



Department of Civil and Environmental Engineering
College of Design and Engineering · National University of Singapore

[**nus.edu.sg/cde**](https://nus.edu.sg/cde)

